Math 0-1: Probability for Data Science & Machine Learning

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1 Probability Basics

To preface this discussion, we're not going to dwell too much on the "definition" of probability in this course, as its main focus is on computation. We looked at 2 different ways of thinking about what probability is. The first way is to think of it as a relative frequency. In this case, the probability of an event occurring is the normalized frequency at which it occurs (i.e. out of 1 or equivalently, 100%). In other words, for some event A:

$$P(A) = \lim_{N \to \infty} \frac{count(A)}{N} \tag{1}$$

Here, N is the number of "experiments" or "trials" that we've done, and A is an event of interest that either occurs or does not occur for each experiment. Note that we would only know the true probability after doing an infinite number of experiments (which of course, we can never do). Thus, probability is more of a "theoretical" measure than one derived by experiment. A common mistake I've seen from past students is that they consider the definition of probability to be the same ratio above, but derived from experimental data. To see why this is wrong, consider an extreme example of flipping a coin once to determine the probability of heads. After only one toss, the result can only be either heads or tails, meaning that you'd assign 100% probability to one outcome and 0% to the other. Intuitively, we know that cannot be true.

The above interpretation of probability doesn't always make sense. As an example, consider when a weather forecaster says something like "there's a 70% chance of rain tomorrow". This does not mean that if we repeated tomorrow an infinite number of times, it would rain 70% of the time. Physical systems are deterministic. It will either rain tomorrow, or it will not. Repeating the same experiment will yield the same result every time.

The second way to think of probability is to think of it as a quantification of uncertainty. In this case, probability is somewhat subjective. But we can still do computation on subjectively derived values.

1.1 Definitions

The sample space Ω is the set of all outcomes of an experiment.

An event is any subset of Ω. This includes Ω itself.

A probability measure P assigns, for each event A, a probability value $P(A)$. In other words, it maps all subsets of Ω to a number. This number is restricted to be in the range $[0, 1]$.

Note that $P(\emptyset) = 0$ and $P(\Omega) = 1$.

1.2 Probability Models

A probability model consists of a nonempty set called a sample space Ω, a collection of events that are subsets of Ω, and a probability measure P assigning a probability between 0 and 1 to each event, with $P(\emptyset) = 0$ and $P(\Omega) = 1$, and where P is additive over disjoint events, i.e. for disjoint events $A_1, A_2, ...$:

$$P(A_1 \cup A2 \cup ...) = P(A_1) + P(A_2) + ... \tag{2}$$

1.3 Properties of Probability Models

Let A^c be the complement of A. i.e. $A^c = \Omega - A$. Then:

$$P(A^c) = 1 - P(A) \tag{3}$$

Let A and B be 2 events.

$$A \subseteq B \implies P(A) \leq P(B) \tag{4}$$

$$A \subseteq B \implies P(B) = P(A) + P(B \cap A^c) \tag{5}$$

$$P(A \cup B) \leq P(A) + P(B) \tag{6}$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \tag{7}$$

1.4 Law of Total Probability

Let $A_1, A_2, ...$ are events that form a partition of the sample space, and B be another event. Then:

$$P(B) = P(A_1 \cap B) + P(A_2 \cap B) + ... \tag{8}$$

This might seem a little random now, but you'll see later in the course how it's applied (joint and marginal distributions).

1.5 Conditional Probability

Let A and B be 2 events. The conditional probability allows us to quantify uncertainty when we have partial information. In this case, the probability of A given that we know B is:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} \tag{9}$$

We can also rearrange to get another commonly used form:

$$P(A \cap B) = P(A \mid B)P(B) \tag{10}$$

This allows us to state the Law of Total Probability in conditional form:

$$P(B) = P(B \mid A_1)P(A_1) + P(B \mid A_2)P(A_2) + ... \tag{11}$$

1.6 Bayes' Rule

By rearranging the conditional probability expressions above, we can express the conditional probability $P(A \mid B)$ in terms of $P(B \mid A)$. i.e. We switch the order of what we know and what is unknown. This has applications in Bayesian machine learning, statistics, etc.

$$P(B \mid A) = \frac{P(A \mid B)P(B)}{P(A)} \tag{12}$$

1.7 Independence

2 events A and B are independent if and only if:

$$P(A \cap B) = P(A)P(B) \tag{13}$$

Equivalently:

$$P(A \mid B) = P(A) \tag{14}$$

And:

$$P(B \mid A) = P(B) \tag{15}$$

A collection of events $A_1, A_2, ...$ are mutually independent if and only if all possible subcollection of events are independent (e.g. $\{A_1, A_2\}$ is a subset of $\{A_1, A_2, A_3\}$).

Note that pairwise independence occurs if all possible pairs of events are independent, but pairwise independence does not imply full mutual independence.

2 Discrete Random Variables and Distributions

A random variable $X(s)$ is a function that assigns a numerical value to each outcome $s \in \Omega$ to a real number. This is useful because it allows us to use mathematical formulas to describe the probability values over a sample space (called a probability distribution).

2.1 Notation

We use $P(X = x)$ to denote a probability distribution. It can be read as "the probability that the random variable X is equal to some specific value x".

We may also use the notation $p_X(x)$ which means the same thing. If it is obvious in context, we may drop the subscript and simply write $p(x)$.

For discrete distributions, we sometimes use letters that are typically used for counting as the argument (e.g. i, j, k, etc.), so we'd have $P(X = k)$ and $p_X(k)$.

$P(X = x) = p_X(x)$ is called the "probability distribution" or "probability mass function" or "PMF". The latter 2 are used only for discrete random variables.

2.2 Bernoulli Distribution

The Bernoulli distribution is the "binary outcome" distribution. Typically we call the 2 outcomes success and failure. Let θ be the probability of success. The support for this distribution is $\{0, 1\}$. The PMF is:

$$p_X(x) = \theta^x (1 - \theta)^{1-x} \tag{16}$$

Alternatively:

$$p_X(x) = \theta x + (1 - \theta)(1 - x) \tag{17}$$

Note: these formulas don't work for x $\notin \{0, 1\}$. We usually won't explicitly state this, but $p_X(x) = 0$ for all values of x outside the support.

Notation:

$$X \sim Bernoulli(\theta) \tag{18}$$

Note: sometimes resources will use p for the probability of success and $q = 1 - p$ for the probability of failure. I will avoid this because we're already using p for the probability distribution.

2.3 Categorical Distribution

The categorical distribution is a more general distribution that assigns a probability value for K different outcomes. K can be 2, but typically we'd just use the Bernoulli for that.

Interesting fact: when using the Bernoulli distribution, we count from 0, but when using the categorical distribution, where there are K possible outcomes, we count from 1.

There isn't any nice formula for the categorical distribution since the probabilities can take on any value (in the support).
But we can say:

$$X \sim Categorical(\theta_1, \theta_2, ..., \theta_K) \tag{19}$$

Or:

$$X \sim Categorical(\vec{\theta}) \tag{20}$$

Here, $\theta_k = P(X = k)$ and $\vec{\theta} = (\theta_1, ..., \theta_K) \in \mathbb{R}^K$.

We also have an additional constraint that:

$$\sum_{k=1}^{K} \theta_k = 1 \tag{21}$$

Using $\mathbb{1}(a)$ as the indicator function (returns 1 when a is true, 0 when a is false), we can write the distribution as:

$$p_X(x) = \theta_1^{\mathbb{1}(x=1)} \theta_2^{\mathbb{1}(x=2)} ... \theta_K^{\mathbb{1}(x=K)} \tag{22}$$

Or:

$$p_X(x) = \prod_{k=1}^{K} \theta_k^{\mathbb{1}(x=k)} \tag{23}$$

Alternatively:

$$p_X(x) = \theta_1 \mathbb{1}(x = 1) + \theta_2 \mathbb{1}(x = 2) + ... + \theta_K \mathbb{1}(x = K) \tag{24}$$

Support:

$$X \in \{1, 2, ..., K\} \tag{25}$$

2.4 Binomial Distribution

The binomial distribution is used for a random variable that represents the number of successes in multiple (n) Bernoulli trials. The PMF is:

$$p_X(x) = \binom{n}{x} \theta^x (1 - \theta)^{n-x} \tag{26}$$

Here, n is the number of trials, θ is the probability of success, and x is the number of successes.

Notation:

$$X \sim Binomial(n, \theta) \tag{27}$$

Support:

$$x \in \{0, 1, 2, ..., n\} \tag{28}$$

2.5 Geometric Distribution

Like the binomial distribution, the geometric distribution is also based on a sequence of Bernoulli trials. But instead of counting the number of successes, it counts the number of failures until a single success. The PMF is:

$$p_X(x) = (1-\theta)^x\theta \tag{29}$$

Notation:

$$X \sim Geometric(\theta) \tag{30}$$

Support:

$$x \in \{0, 1, 2, ...\} \tag{31}$$

2.6 Poisson Distribution

The Poisson distribution is typically used for modeling counts, but we derived it as a limit of the binomial distribution (where we assumed that the probability of success is very small, so that a large n is required to see a success). The PMF is:

$$p_X(x) = \frac{\lambda^x}{x!}e^{-\lambda} \tag{32}$$

Notation:

$$X \sim Poisson(\lambda) \tag{33}$$

Support:

$$x \in \{0, 1, 2, ...\} \tag{34}$$

3 Continuous Random Variables and Distributions

In our exploration of discrete random variables, we have focused on situations where outcomes are countable—like the roll of a die or the number of heads in a series of coin flips. These variables take on distinct, separate values, each with a corresponding probability. This framework is useful for many practical problems, especially those involving well-defined, finite sets of outcomes.

However, not all random phenomena fit neatly into the mold of countable outcomes. What if we wanted to describe the amount of time until a light bulb burns out, or the exact height of a tree? Here, the possible outcomes are not isolated points but form a continuum—an infinite range of values within a given interval. This is where continuous random variables come into play.

Unlike discrete random variables, which are characterized by distinct, separate outcomes, continuous random variables can take any value within a specified range. Instead of assigning probabilities to specific outcomes, we deal with probabilities over intervals, necessitating a shift from probability mass functions to probability density functions. This transition introduces new mathematical tools, such as integration, that allow us to describe and analyze the behavior of these variables.

Interesting fact about continuous values: it doesn't matter how small a range of values we look at. There are still an infinite number of values in that range! This should help give you some intuition that, for a continuous random variable X, $P(X = x) = 0, \forall x \in \mathbb{R}$.

We'll use the notation $f_X(x)$ to represent the probability density function (PDF) of a random variable X. In future / more advanced courses, we may simply use $p(x)$.

3 properties of PDFs:

$$f_X(x) \geq 0 \tag{35}$$

$$P(a \leq X \leq b) = \int_a^b f_X(x)dx \tag{36}$$

$$\int_{-\infty}^{\infty} f_X(x)dx = 1 \tag{37}$$

3.1 Uniform Distribution

This distribution has the same density for all values in the support. This is an important distribution because in programming libraries, it's the "base" distribution from which samples of other distributions can be drawn. In other words, we have some formula to draw samples from the uniform distribution. Then, if we want samples from some other distribution, e.g. the exponential or the normal, we can transform that sample appropriately so that the result comes from the desired distribution.

$$f_X(x) = \frac{1}{b-a}\mathbb{1}(a \leq x \leq b) \tag{38}$$

Notation:

$$X \sim U(a, b) \tag{39}$$

Sometimes we use a fancy "U":

$$X \sim \mathcal{U}(a, b) \tag{40}$$

Support:

$$x \in [a, b] \tag{41}$$

Parameter Constraints:

$$a < b \tag{42}$$

3.2 Exponential Distribution

λ is called the "rate parameter". This alludes to the application of this distribution in the real-world, which is to model "inter-arrival" times. E.g. modeling when a bus will arrive, when a customer will walk into the store, or when a user will visit your website.

$$f_X(x) = \lambda e^{-\lambda x}\mathbb{1}(x \geq 0) \tag{43}$$

Notation:

$$X \sim Exp(\lambda) \tag{44}$$

Support:

$$x \in [0, \infty) \tag{45}$$

Parameter Constraints:

$$\lambda > 0 \tag{46}$$

3.3 Normal / Gaussian Distribution

The famous "bell curve". Used essentially everywhere, any time there is a continuous value to model. The parameters of the distribution are very important quantities. μ is called the mean and σ^2 is called the variance. The mean tells us where the "center" of the distribution is, while the variance tells us how spread out the distribution is. We'll learn more about these concepts later when we discuss expectation.

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}\frac{(x-\mu)^2}{\sigma^2}} \tag{47}$$

Notation:

$$X \sim N(\mu, \sigma^2) \tag{48}$$

Sometimes we use a fancy "N":

$$X \sim \mathcal{N}(\mu, \sigma^2) \tag{49}$$

Support:

$$x \in (-\infty, \infty) \tag{50}$$

Parameter Constraints:

$$\mu \in \mathbb{R}, \sigma^2 > 0 \tag{51}$$

3.4 Laplace Distribution

Not usually discussed in a typical probability course, but appears often enough in my courses to warrant a mention. Also called the "double exponential" and allows for another parameter to control the centre of the distribution.

$$f_X(x) = \frac{1}{2b} e^{\frac{-|x-a|}{b}} \tag{52}$$

Notation:

$$X \sim Laplace(a, b) \tag{53}$$

Support:

$$x \in (-\infty, \infty) \tag{54}$$

4 More About Random Variables and Distributions

4.1 Cumulative Distribution Function (CDF)

We define the CDF as:

$$F_X(x) = P(X \leq x) \tag{55}$$

In the discrete case:

$$F_X(x) = \sum_{k=-\infty}^{x} p_X(k) \tag{56}$$

In the continuous case:

$$F_X(x) = \int_{-\infty}^{x} f_X(t)dt \tag{57}$$

And due to the Fundamental Theorem of Calculus, we also have that:

$$f_X(x) = \frac{dF_X(x)}{dx} \tag{58}$$

Key point: you can always determine the full distribution (i.e. PMF or PDF) of a random variable from its CDF.

4.2 Properties of Cumulative Distribution Function (CDF)

$$0 \leq F_X(x) \leq 1 \tag{59}$$

$$x \leq y \implies F_X(x) \leq F_X(y) \tag{60}$$

$$\lim_{x \to +\infty} F_X(x) = 1 \tag{61}$$

$$\lim_{x \to -\infty} F_X(x) = 0 \tag{62}$$

4.3 Functions of Random Variables / Change of Variables

If X is a discrete random variable, $h : \mathbb{R} \to \mathbb{R}$, and $Y = h(X)$ is also a discrete random variable, then:

$$p_Y(y) = \sum_{x \in h^{-1}(y)} p_X(x) \tag{63}$$

Where $h^{-1}(y)$ returns a set that contains all the values of x that would produce a given value of y.

If X is continuous, and h is such that $Y = h(X)$ is discrete, then:

$$p_Y(y) = \int_{x \in h^{-1}(y)} f_X(x)dx \tag{64}$$

If X is continuous, $Y = h(X)$ is continuous, and h is a 1-to-1 function, then:

$$f_Y(y) = \frac{f_X(h^{-1}(y))}{|h'(h^{-1}(y)|} \tag{65}$$

If h is not 1-to-1, then you must sum over all possible values of x that would lead to the given y, as above.

4.4 Joint and Marginal Distributions

Suppose we have 2 random variables X and Y. The joint CDF is:

$$F_{X,Y}(x, y) = P(X \leq x, Y \leq y) \tag{66}$$

The marginal CDF is given by:

$$F_X(x) = \lim_{y \to \infty} F_{X,Y}(x, y) \tag{67}$$

Similarly:

$$F_Y(y) = \lim_{x \to \infty} F_{X,Y}(x, y) \tag{68}$$

The joint PMF for discrete X and Y is described as:

$$p_{X,Y}(x, y) = P(X = x, Y = y) \tag{69}$$

The marginal PMF is given by:

$$p_X(x) = \sum_{y=-\infty}^{\infty} p_{X,Y}(x, y) \tag{70}$$

For a continuous X and Y, we have that:

$$p_{X,Y}(x,y) = 0, \forall x, y \in \mathbb{R}^2 \tag{71}$$

The joint PDF has the constraint:

$$f_{X,Y}(x,y) \geq 0, \forall x, y \in \mathbb{R}^2 \tag{72}$$

Properties:

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f_{X,Y}(x,y)dxdy = 1 \tag{73}$$

The marginal PDF is given by:

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)dy \tag{74}$$

And similarly for Y.

A special joint distribution we'll be working with quite a bit is the bivariate normal. It has the PDF:

$$f_{X,Y}(x,y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left\{-\frac{1}{2(1-\rho^2)}\left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2 + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2 - 2\rho\left(\frac{x-\mu_X}{\sigma_X}\right)\left(\frac{y-\mu_Y}{\sigma_Y}\right)\right]\right\} \tag{75}$$

4.5 Conditional Distributions and Bayes' Rule

For discrete random variables X and Y, we have:

$$p_{Y|X}(y \mid x) = P(Y = y \mid X = x) = \frac{p_{X,Y}(x,y)}{p_X(x)} = \frac{p_{X,Y}(x,y)}{\sum_{y'} p_{X,Y}(x,y')} \tag{76}$$

And Bayes' rule can be stated as:

$$p_{Y|X}(y \mid x) = \frac{p_{X|Y}(x \mid y)p_Y(y)}{p_X(x)} = \frac{p_{X|Y}(x \mid y)p_Y(y)}{\sum_{y'} p_{X|Y}(x \mid y')p_Y(y')} \tag{77}$$

For continuous random variables X and Y, we have the same thing except we replace p with f:

$$f_{Y|X}(y \mid x) = \frac{f_{X,Y}(x,y)}{f_X(x)} = \frac{f_{X,Y}(x,y)}{\sum_{y'} f_{X,Y}(x,y')} \tag{78}$$

And Bayes' rule can be stated as:

$$f_{Y|X}(y \mid x) = \frac{f_{X|Y}(x \mid y)f_Y(y)}{f_X(x)} = \frac{f_{X|Y}(x \mid y)f_Y(y)}{\sum_{y'} f_{X|Y}(x \mid y')f_Y(y')} \tag{79}$$

4.6 Independence

$X \perp\!\!\!\perp Y$ means "X is independent of Y".

For discrete random variables:

$$X \perp\!\!\!\perp Y \iff p_{X,Y}(x,y) = p_X(x)p_Y(y) \tag{80}$$

$$X \perp\!\!\!\perp Y \iff p_{X|Y}(x \mid y) = p_X(x) \tag{81}$$

For continuous random variables:

$$X \perp\!\!\!\perp Y \iff f_{X,Y}(x,y) = f_X(x)f_Y(y) \tag{82}$$

$$X \perp\!\!\!\perp Y \iff f_{X|Y}(x \mid y) = f_X(x) \tag{83}$$

If 2 random variables are independent, then functions of those random variables are also independent:

$$X \perp\!\!\!\perp Y \implies g(X) \perp\!\!\!\perp h(Y) \tag{84}$$

For more than 2 discrete random variables, we say that $X_1, X_2, ..., X_n$ are jointly independent (or simply "independent"), if and only if:

$$p_{X_1,X_2,...,X_n}(x_1, x_2, ..., x_n) = p_{X_1}(x_1)p_{X_2}(x_2)...p_{X_n}(x_n) \tag{85}$$

And for continuous random variables:

$$f_{X_1,X_2,...,X_n}(x_1, x_2, ..., x_n) = f_{X_1}(x_1)f_{X_2}(x_2)...f_{X_n}(x_n) \tag{86}$$

"Independent and identically distributed" (a.k.a. iid) means that $X_1, X_2, ..., X_n$ are jointly independent and all come from the same distribution.

One scenario where this happens is when we have drawn n "samples" from a distribution.

4.7 Multivariate Distributions

Suppose we have a collection of n random variables, $X_1, X_2, ..., X_n$.

If they are continuous, the joint distribution integrates to 1:

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}...\int_{-\infty}^{\infty} f_{X_1,X_2,...,X_n}(x_1, x_2, ..., x_n)dx_1dx_2...dx_n = 1 \tag{87}$$

And it can be used to find the probability that $X_1, X_2, ..., X_n$ is within a certain volume:

$$P(a_1 \leq X_1 \leq b_1, a_2 \leq X_2 \leq b_2, ..., a_n \leq X_n \leq b_n) = \int_{a_1}^{b_1}\int_{a_2}^{b_2}...\int_{a_n}^{b_n} f_{X_1,X_2,...,X_n}(x_1, x_2, ..., x_n)dx_1dx_2...dx_n \tag{88}$$

If they are discrete, then the joint distribution sums to 1:

$$\sum_{x_1=-\infty}^{\infty}\sum_{x_2=-\infty}^{\infty}...\sum_{x_n=-\infty}^{\infty} p_{X_1,X_2,...,X_n}(x_1, x_2, ..., x_n) = 1 \tag{89}$$

And it can be used to find the probability that $X_1, X_2, ..., X_n$ is within a certain volume:

$$P(a_1 \leq X_1 \leq b_1, a_2 \leq X_2 \leq b_2, ..., a_n \leq X_n \leq b_n) = \sum_{x_1=a_1}^{b_1}\sum_{x_2=a_2}^{b_2}...\sum_{x_n=a_n}^{b_n} p_{X_1,X_2,...,X_n}(x_1, x_2, ..., x_n) \tag{90}$$

A random vector is a vector of random variables. Let $\boldsymbol{X} = (X_1, X_2, ..., X_n)^T \in \mathbb{R}^n$.

We'll write $f_{\boldsymbol{X}}(\boldsymbol{x})$ for the PDF and $p_{\boldsymbol{X}}(\boldsymbol{x})$ for the PMF.

Note that the rules we learned for 2 random variables also apply for 2 random vectors (e.g. conditional distributions, independence, Bayes' rule).

For continuous random variables, we can marginalize over a random vector as follows:

$$f_{\boldsymbol{X}}(\boldsymbol{x}) = \int f_{\boldsymbol{X},\boldsymbol{Y}}(\boldsymbol{x}, \boldsymbol{y})d\boldsymbol{y} \tag{91}$$

We can compute a conditional distribution as follows:

$$f_{\boldsymbol{Y}|\boldsymbol{X}}(\boldsymbol{y} \mid \boldsymbol{x}) = \frac{f_{\boldsymbol{X},\boldsymbol{Y}}(\boldsymbol{x}, \boldsymbol{y})}{f_{\boldsymbol{X}}(\boldsymbol{x})} \tag{92}$$

And Bayes' rule looks as follows:

$$f_{\boldsymbol{Y}|\boldsymbol{X}}(\boldsymbol{y} \mid \boldsymbol{x}) = \frac{f_{\boldsymbol{X}|\boldsymbol{Y}}(\boldsymbol{x} \mid \boldsymbol{y}) f_{\boldsymbol{Y}}(\boldsymbol{y})}{f_{\boldsymbol{X}}(\boldsymbol{x})} \tag{93}$$

For discrete random variables, we can marginalize over a random vector as follows:

$$p_{\boldsymbol{X}}(\boldsymbol{x}) = \sum_{\boldsymbol{y}} p_{\boldsymbol{X},\boldsymbol{Y}}(\boldsymbol{x}, \boldsymbol{y}) \tag{94}$$

We can compute a conditional distribution as follows:

$$p_{\boldsymbol{Y}|\boldsymbol{X}}(\boldsymbol{y} \mid \boldsymbol{x}) = \frac{p_{\boldsymbol{X},\boldsymbol{Y}}(\boldsymbol{x}, \boldsymbol{y})}{p_{\boldsymbol{X}}(\boldsymbol{x})} \tag{95}$$

And Bayes' rule looks as follows:

$$p_{\boldsymbol{Y}|\boldsymbol{X}}(\boldsymbol{y} \mid \boldsymbol{x}) = \frac{p_{\boldsymbol{X}|\boldsymbol{Y}}(\boldsymbol{x} \mid \boldsymbol{y}) p_{\boldsymbol{Y}}(\boldsymbol{y})}{p_{\boldsymbol{X}}(\boldsymbol{x})} \tag{96}$$

2 random vectors are considered independent if and only if their joint distribution factorizes (note that the distributions over each component of the vectors don't have to factorize).

For discrete random variables:

$$\boldsymbol{X} \perp\!\!\!\perp \boldsymbol{Y} \iff p_{\boldsymbol{X},\boldsymbol{Y}}(\boldsymbol{x}, \boldsymbol{y}) = p_{\boldsymbol{X}}(\boldsymbol{x}) p_{\boldsymbol{Y}}(\boldsymbol{y}) \tag{97}$$

$$\boldsymbol{X} \perp\!\!\!\perp \boldsymbol{Y} \iff p_{\boldsymbol{X}|\boldsymbol{Y}}(\boldsymbol{x} \mid \boldsymbol{y}) = p_{\boldsymbol{X}}(\boldsymbol{x}) \tag{98}$$

For continuous random variables:

$$\boldsymbol{X} \perp\!\!\!\perp \boldsymbol{Y} \iff f_{\boldsymbol{X},\boldsymbol{Y}}(\boldsymbol{x}, \boldsymbol{y}) = f_{\boldsymbol{X}}(\boldsymbol{x}) f_{\boldsymbol{Y}}(\boldsymbol{y}) \tag{99}$$

$$\boldsymbol{X} \perp\!\!\!\perp \boldsymbol{Y} \iff f_{\boldsymbol{X}|\boldsymbol{Y}}(\boldsymbol{x} \mid \boldsymbol{y}) = f_{\boldsymbol{X}}(\boldsymbol{x}) \tag{100}$$

Note that in the above cases, distributions like $f_{\boldsymbol{X}}(\boldsymbol{x})$ are being referred to as "marginals", but they are also "joint" distributions over the components of $\boldsymbol{X}$.

In future courses and other courses (outside of my own), you may find that we won't bother to bold symbols or put arrows on top of them. Conveniently, all the rules look exactly the same in both vector and scalar form.

4.8 Multivariate Normal

Used for modeling continuous multidimensional vectors. PDF for a random vector in $\mathbb{R}^D$:

$$f_{\boldsymbol{X}}(\boldsymbol{x}) = \frac{1}{\sqrt{(2\pi)^D |\Sigma|}} e^{-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^T \Sigma^{-1} (\boldsymbol{x}-\boldsymbol{\mu})} \tag{101}$$

Here, the mean vector $\boldsymbol{\mu} \in \mathbb{R}^D$ and the covariance matrix $\Sigma \in \mathbb{R}^{D \times D}$ is positive semi-definite.

Note that $\Sigma_{ii} = \sigma_i^2$, the variance of component X_i.

Notation:

$$\boldsymbol{X} \sim N(\boldsymbol{\mu}, \Sigma) \tag{102}$$

Relationship between correlation and covariance: Let ρ_{ij} be the correlation between X_i and X_j, and let Σ_{ij} be the covariance between X_i and X_j. Then:

$$\Sigma_{ij} = \rho_{ij} \sigma_i \sigma_j \tag{103}$$

Here σ_i is the standard deviation (square root of variance) of X_i.

Support:

$$\boldsymbol{x} \in \mathbb{R}^D \tag{104}$$

4.9 Multinomial Distribution

Arises from "rolling a K-sided die" multiple times. Analogous to going from Bernoulli to binomial. This is the "multiple trials" version of the categorical.

There are $K+1$ parameters, corresponding to $\boldsymbol{\theta} = (\theta_1, ..., \theta_K)^T$, the probability of choosing $X_1, ..., X_K$ on each trial, and n, the number of trials.

Notation:

$$\boldsymbol{X} \sim Multinomial(n, \boldsymbol{\theta}) \tag{105}$$

The PMF looks as follows:

$$p_{\boldsymbol{X}}(\boldsymbol{x}) = \frac{n!}{x_1! x_2! ... x_K!} \theta_1^{x_1} \theta_2^{x_2} ... \theta_K^{x_K} \tag{106}$$

Support:

$$\boldsymbol{x} \in \{\boldsymbol{x} \mid x_j \in \mathbb{N}_0, \sum_{j=1}^{K} x_j = n\} \tag{107}$$

Here $\mathbb{N}_0 = \{0, 1, 2, ...\}$.

4.10 Multidimensional Change of Variables

Let $\boldsymbol{X}$ and $\boldsymbol{Y}$ be discrete random vectors of the same size, and $\boldsymbol{Y} = h(\boldsymbol{X})$. Then:

$$p_{\boldsymbol{Y}}(\boldsymbol{y}) = \sum_{\boldsymbol{x} \in h^{-1}(\boldsymbol{y})} p_{\boldsymbol{X}}(\boldsymbol{x}) \tag{108}$$

Here, $h^{-1}(\boldsymbol{y})$ returns a set of all $\boldsymbol{x}$ that would yield $\boldsymbol{y}$ when passed through the function h.

Let $\boldsymbol{X}$ and $\boldsymbol{Y}$ be continuous random vectors of the same size, and $\boldsymbol{Y} = h(\boldsymbol{X})$, and h be a 1-to-1 function. Then:

$$f_{\boldsymbol{Y}}(\boldsymbol{y}) = \frac{f_{\boldsymbol{X}}(h^{-1}(\boldsymbol{y}))}{abs(det(J))} \tag{109}$$

Where J is the Jacobian matrix of partial derivatives of h:

$$J = \frac{\partial \boldsymbol{y}}{\partial \boldsymbol{x}} \tag{110}$$

Here, I've explicitly stated the absolute value and determinant operations since they both use the same math symbol.

4.11 Convolution

Let X and Y be independent random variables. The sum $Z = X + Y$ is also a random variable.

If X and Y are discrete:

$$p_Z(z) = \sum_w p_X(z-w) p_Y(w) \tag{111}$$

If X and Y are continuous:

$$f_Z(z) = \int f_X(z-w) f_Y(w) dw \tag{112}$$

5 Expectation

5.1 Mean

The expected value of a random variable X is denoted as $E(X)$. There are many other ways to write this, including $E[X]$, $\mathbb{E}[X]$, $\mathbb{E}X$, $\langle X \rangle$, etc. We call E the "expectation operator".

For discrete random variables:

$$E(X) = \sum_{x=-\infty}^{\infty} x p_X(x) \tag{113}$$

For continuous random variables:

$$E(X) = \int_{-\infty}^{\infty} x f_X(x) dx \tag{114}$$

The expected value is also known as the "mean". This tells us the "center of mass" of a distribution. Another way to think about the mean is that it's the "average". So $E(X)$ tells us the "average value of X".
We'll often use the symbol $\mu = E(X)$.

5.2 Properties of Expectation

The expected value of a function $g(X)$ is (for continuous X):

$$E(g(X)) = \int_{-\infty}^{\infty} g(x) f_X(x) dx \tag{115}$$

Or (for discrete X):

$$E(g(X)) = \sum_{x=-\infty}^{\infty} g(x) p_X(x) \tag{116}$$

Note: also works for vector functions and vector random variables.

Expectation is a linear operator. So:

$$E(aX + bY) = aE(X) + bE(Y) \tag{117}$$

If X and Y are independent, their expected values factorize:

$$X \perp\!\!\!\perp Y \implies E(XY) = E(X)E(Y) \tag{118}$$

5.3 Variance

Variance is defined as follows:

$$Var(X) = E\left[(X - \mu)^2\right] \tag{119}$$

Recall that $\mu = E(X)$.

So in other words, the variance of a random variable is the "average of the square of the deviation of the random variable from its mean". Put another way, it tells us, on average, how far away X is from its mean. In other words, "how spread out" the distribution is.

We can manipulate the expression for variance to get another form that is often useful:

$$Var(X) = E\left[X^2\right] - E\left[X\right]^2 = E\left[X^2\right] - \mu^2 \tag{120}$$

Additionally, note that we often use the notation $\sigma^2 = Var(X)$. And moreover, we call σ (the positive square root of the variance) the "standard deviation".

5.4 Moments, Skewness, Kurtosis

We can consider even higher powers of X inside the expected value.

We call $E(X^n)$ the nth moment of X.

Skewness is the third standardized moment:

$$Sk(X) = E\left[\left(\frac{x-\mu}{\sigma}\right)^3\right] \tag{121}$$

Kurtosis is the fourth standardized moment.

$$Kur(X) = E\left[\left(\frac{x-\mu}{\sigma}\right)^4\right] \tag{122}$$

Skewness tells us how asymmetric the distribution is, and in which direction. Kurtosis tells us how heavy the tails of the distribution are.

The normal distribution has a kurtosis of 3. Because of this, we often speak of the "excess kurtosis", which is given by how much larger the kurtosis is relative to 3.

5.5 Covariance and Correlation

The covariance between 2 random variables X and Y is:

$$Cov(X,Y) = E\left[(X-\mu_X)(Y-\mu_Y)\right] \tag{123}$$

The correlation between 2 random variables X and Y is:

$$Corr(X,Y) = \frac{Cov(X,Y)}{\sigma_X \sigma_Y} \tag{124}$$

Note: correlation is always between -1 and +1. Think of it as "normalized covariance".

Important facts:

$$Corr(X,Y) = 0 \not\Rightarrow X \perp\!\!\!\perp Y \tag{125}$$

Zero correlation / covariance does not imply independence.

$$X \perp\!\!\!\perp Y \implies Corr(X,Y) = 0 \tag{126}$$

Independence does imply zero correlation / covariance.

Note: correlation measures linear dependence. To demonstrate this, we looked at an example where if Y is a linear function of X, then their correlation is 1. Conversely, we looked at an example earlier where Y is a nonlinear function of X, so that Y is completely determined by X, but their correlation was 0.

5.6 Covariance and Correlation Matrices

Given the random vector $\boldsymbol{X}$, its mean vector is $\boldsymbol{\mu} = E(\boldsymbol{X})$, and its covariance matrix is:

$$\Sigma = E\left[(\boldsymbol{X}-\boldsymbol{\mu})(\boldsymbol{X}-\boldsymbol{\mu})^T\right] \tag{127}$$

The elements of the correlation matrix are given by:

$$\rho_{ij} = \frac{\Sigma_{ij}}{\sigma_i \sigma_j} \tag{128}$$

5.7 Conditional Expectation

Discrete case:

$$E(X \mid Y = y) = \sum_x x p_{X|Y}(x \mid y) \tag{129}$$

Continuous case:

$$E(X \mid Y = y) = \int x f_{X|Y}(x \mid y) dx \tag{130}$$

We can also speak of $E(X \mid Y)$, which considers Y as a random variable not being set to a specific value of y. Because this is a function of the random variable Y, this quantity is also a random variable!

Conditional expectation is linear.

$$E(aX_1 + bX_2 \mid Y) = aE(X_1 \mid Y) + bE(X_2 \mid Y) \tag{131}$$

5.8 Law of Total Expectation

We previously discussed that $E(X \mid Y)$ is a random variable, so we can ask "what is its expected value?"

The law of total expectation says:

$$E(E(X \mid Y)) = E(X) \tag{132}$$

Application: reinforcement learning (Bellman equation derivation).

6 Generating Functions

We define the Moment Generating Function or MGF for the random variable X as:

$$M_X(t) = E(e^{tX}) \tag{133}$$

The MGF does not always exist. The Characteristic Function or CF, always exists, but requires us to work with complex variables. It is defined as:

$$\varphi_X(t) = E(e^{itX}) \tag{134}$$

The MGF of a random vector $\boldsymbol{X}$ is:

$$M_{\boldsymbol{X}}(\boldsymbol{t}) = E(e^{\boldsymbol{t}^T \boldsymbol{X}}) \tag{135}$$

The CF of a random vector $\boldsymbol{X}$ is:

$$M_{\boldsymbol{X}}(\boldsymbol{t}) = E(e^{i\boldsymbol{t}^T \boldsymbol{X}}) \tag{136}$$

If we have multiple independent random variables $X_1, X_2, ..., X_n$, then the MGF of their sum is:

$$M_{X_1+...+X_n}(t) = E(e^{t(X_1+...+X_n)}) = E(e^{tX_1})...E(e^{tX_n}) = M_{X_1}(t)M_{X_2}(t)...M_{X_n}(t) \tag{137}$$

Furthermore, if they are iid with the same distribution as some random variable X, then:

$$M_{X_1+...+X_n}(t) = (M_X(t))^n \tag{138}$$

As suggested by its name, the MGF can be used to find the moments of a distribution. In particular:

$$M_X^{(k)}(0) = E(X^k) \tag{139}$$

Similarly, the CF can also be used to find the moments of a distribution. In particular:

$$\varphi_X^{(k)}(0) = i^k E(X^k) \tag{140}$$

Uniqueness theorem: If X and Y have the same characteristic function (or moment generating function), then they have the same distribution.

7 Inequalities

We first introduced the monotonicity property of expectation:

$$X \leq Y \implies E(X) \leq E(Y) \tag{141}$$

We also looked at the non-negativity property:

$$X \geq 0 \implies E(X) \geq 0 \tag{142}$$

We then looked at the Markov inequality. For a random variable X that is non-negative, and any value $a > 0$, we have:

$$P(X \geq a) \leq \frac{E(X)}{a} \tag{143}$$

Using the Markov inequality, we derived the Chebyshev inequality, which has many applications (including in this course). It states that for a random variable Y with finite mean μ, and for any value $a > 0$, we have:

$$P(|Y - \mu| \geq a) \leq \frac{Var(Y)}{a^2} \tag{144}$$

We then looked at the Cauchy-Schwartz inequality, which we used to prove that correlation is always between -1 and +1. For 2 random variables X and Y, it states:

$$|Cov(X, Y)| \leq \sqrt{Var(X)Var(Y)} \tag{145}$$

8 Limit Theorems

In this section we introduce 3 types of convergence. This is for practical reasons; namely that the law of large numbers and central limit theorem are examples of these types of convergence.

The first type of convergence we discussed was "convergence in probability". A sequence of random variables $\{X_n\} = \{X_1, X_2, ...\}$ converges in probability if:

$$\lim_{n\to\infty} P(|X_n - Y| \geq \epsilon) = 0 \tag{146}$$

For all $\epsilon > 0$. Here, Y is also another random variable, but also can be a constant.

Notation for convergence in probability:

$$X_n \xrightarrow{P} Y \tag{147}$$

The "weak law of large numbers" (WLLN) is an example of convergence in probability. It states that, for an infinite sequence $\{X_n\} = \{X_1, X_2, ...\}$ of independent random variables with the same mean μ and finite variance, the sample mean converges in probability to the true mean. i.e. If we let the sample mean be M_n:

$$M_n = \frac{1}{n}\sum_{i=1}^{n} X_i \tag{148}$$

Then:

$$\lim_{n\to\infty} P(|M_n - \mu| \geq \epsilon) = 0 \tag{149}$$

We then looked at convergence with probability 1 (a.k.a. "almost sure" convergence). We say that the sequence "converges almost surely". A sequence of random variables $\{X_n\} = \{X_1, X_2, ...\}$ converges with probability 1 to another random variable Y (which might be constant), if:

$$P\left(\lim_{n\to\infty} X_n = Y\right) = 1 \tag{150}$$

The notation for almost sure convergence is:

$$X_n \overset{a.s.}{\to} Y \tag{151}$$

The "strong law of large numbers" (SLLN) is an example of almost sure convergence. Here, let $\{X_n\} = \{X_1, X_2, ...\}$ be a sequence of iid random variables with finite mean μ (note: this condition is more strict than for the WLLN) (note 2: there is no restriction on variance, it can be infinite). Then, the SLLN states that:

$$P\left(\lim_{n\to\infty} M_n = \mu\right) = 1 \tag{152}$$

Finally, we looked at convergence in distribution. Let $\{X, X_1, X_2, ...\}$ be random variables. The sequence $\{X_n\}$ converges in distribution to X, if, for all $x \in \mathbb{R}$ such that $P(X = x) = 0$, we have:

$$\lim_{n\to\infty} P(X_n \leq x) = P(X \leq x) \tag{153}$$

Convergence in distribution is denoted:

$$X_n \overset{d}{\to} X \tag{154}$$

Some types of convergence imply that others are also true. For the types of convergence we discussed, the ordering is as follows:

$$X_n \overset{a.s.}{\to} Y \implies X_n \overset{P}{\to} X \implies X_n \overset{d}{\to} X \tag{155}$$

Here, anything on the left implies everything on the right.

The central limit theorem (CLT) is an example of convergence in distribution. The CLT states that, for a sequence of iid random variables $\{X_n\} = \{X_1, X_2, ...\}$ with finite mean μ and finite variance σ^2, the standardized sample mean:

$$Z_n = \frac{M_n - \mu}{\sigma/\sqrt{n}} \tag{156}$$

converges in distribution to the standard normal distribution. i.e.:

$$Z_n \overset{d}{\to} Z \tag{157}$$

where $Z \sim \mathcal{N}(0, 1)$.

In other words, sums (or equivalently, sample means) of iid random variables with finite mean and variance approach a normal distribution. This is quite useful for many approximations done in statistics and beyond.

9 Other Topics

9.1 Chain Rule of Probability

The chain rule (nothing to do with calculus) of probability states that, for a set of random variables $X_1, X_2, ..., X_n$, we can factorize its joint distribution as follows:

$$p(x_1, x_2, ..., x_n) = p(x_1)p(x_2 \mid x_1)p(x_3 \mid x_1, x_2)...p(x_n \mid x_1, x_2, ..., x_{n-1}) \tag{158}$$